

Math 571 - Homework 6

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Problem 6.1 (R:6.1). Suppose $\alpha : [a, b] \rightarrow \mathbb{R}$ is monotonic increasing and continuous at $x_0 \in [a, b]$. consider $f : [a, b] \rightarrow \{0, 1\}$ given by $f(x_0) = 1$ and $f(x) = 0$ for $x \neq x_0$. Show that $f \in \mathcal{R}(\alpha)$ and $\int_a^b f d\alpha = 0$.

Problem 6.2 (R:6.2). Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous, $f \geq 0$, and $\int_a^b f dx = 0$, then $f = 0$.

Note that where Rudin asks you to compare with (R:6.1). You might think that these are not comparable since (R:6.1) is about $\mathcal{R}(\alpha)$ while (R:6.2) is about $\mathcal{R}$, but taking $\alpha = \text{id}$ in (R:6.1) allows you to make the comparison.

Problem 6.3 (R:6.3). Define $\beta_i : [-1, 1] \rightarrow [0, 1]$ by $\beta_i = 0$ for $x < 0$ and $\beta_i = 1$ for $x > 0$, then $\beta_1(0) = 0$, $\beta_2(0) = 1$, and $\beta_3(0) = 1/2$. In particular β_i has a simple discontinuity at 0 with $\beta_1(0-) = \beta_1(0) = 0$ (continuous from the left), $\beta_2(0+) = \beta_2(0) = 1$ (continuous from the right), while β_3 is neither continuous from the left or right. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be bounded. show that

- (a) $f \in \mathcal{R}(\beta_1)$ iff $f(0+) = f(0)$, that is, f is continuous from the right at 0.
- (b) $f \in \mathcal{R}(\beta_2)$ iff $f(0-) = f(0)$, that is, f is continuous from the left at 0.
- (c) $f \in \mathcal{R}(\beta_3)$ iff f is continuous at 0.
- (d) Given an interval $[a, b]$ and a point $c \in [a, b]$, let $\beta_{j,a,b,c}$ ($j = 1, 2$) be the functions on $[a, b]$ which equal 0 on $[a, c)$, 1 on $(c, b]$, and are 0 or 1 at c depending on whether j is 1 or 2. (So for $j = 1, 2$, Rudin's β_j is $\beta_{j,-1,1,0}$ in this notation.) **State the results analogous to Rudin's (a) and (b) for integrals $\int_a^b f(x) d\beta_{j,a,b,c}(x)$.** Do not hand in proofs. But think things through carefully; you will be graded on the correctness of your answers. The cases $c = a$ and $c = b$ are slightly different from the general case, so I recommend that you first state the results for $c \in (a, b)$, then say how they must be modified for those cases.
- (e) Assuming the results of part (e), deduce that a function f on an interval $[a, b]$ is integrable with respect to *every* increasing function α if and only if it is continuous.

Problem 6.4 (B:3.3:3 – An α which does all its increasing on the Cantor set.). Let $\alpha : [0, 1] \rightarrow [0, 1]$ be the Cantor function from Problem 4.10 of Homework 4. The main fact about α is that it is an increasing (non-decreasing) continuous function from $[0, 1]$ onto $[0, 1]$ that is absolutely flat off the Cantor set. So α only increases on a measure 0 set, yet is surjective!

- (a) Show that if f is a continuous real-valued function on $[0, 1]$ which is zero on all points of the Cantor set, then $\int f d\alpha = 0$.
- (b) Give an example of a function f as in (a) which is not the zero function.
- (c) Deduce from (a) that if f and g are two continuous real-valued functions on $[0, 1]$ which agree on all points of the Cantor set, then $\int f d\alpha = \int g d\alpha$.
- (d) Let $f : [0, 1] \rightarrow \mathbb{R}$ be the function such that $f(x) = 0$ for all x in the Cantor set, and $f(x) = 1$ for all other x . Show that $f \notin \mathcal{R}(\alpha)$.
- (e) Why does the result of (d) not contradict the result of (a) above? Why does it not contradict the result of **R:6.6**?

Problem 6.5 (B:6.4:9 – Functions with only countable many discontinuities are integrable.). Let f be bounded on $[a, b]$ with at most countable many discontinuities on $[a, b]$. Let $\alpha : [a, b] \rightarrow \mathbb{R}$ is monotonic increasing and α is continuous at every discontinuity of f . Show that $f \in \mathcal{R}(\alpha)$.

Hint: Fix an enumeration $S = \{s_i \mid i \in \mathbb{N}\}$ of the discontinuities of f . Fix $\epsilon > 0$ and $\epsilon_i > 0$ so that $\sum_i \epsilon_i \leq \epsilon$. Since α is continuous at s_i fix δ_i so that $\alpha(N_{\delta_i}(s_i)) \subset N_{\epsilon_i}(\alpha(s_i))$. For $x \notin S$, fix δ_x so that $f(N_{\delta_x}(x)) \subset N_\epsilon(f(x))$. Now $\mathcal{O} = \{N_{\delta_i}(s_i) \mid i \in \mathbb{N}\} \cup \{N_{\delta_x}(x) \mid x \notin S\}$ is an open cover of $[a, b]$. Apply compactness to get a finite subcover and then do something *similar* (not the same) as in the proof of 6.10.

Problem 6.6 (R:6:6 or B:6.4:10 – A Riemann Integrable function with uncountably many discontinuities.). Let $f : [0, 1] \rightarrow \mathbb{R}$ be bounded and continuous off of the Cantor set $\mathcal{C}$. Show that $f \in \mathcal{R}$.

Summary of Problems 6.4 - 6.6 and a generalization: Let $\text{Dis}(f) = \{x \mid x \text{ is discontinuous at } x\}$. Let $\alpha : [0, 1] \rightarrow [0, 1]$ be the Cantor function (“*devil’s staircase*”) and let $f = \chi_{\mathcal{C}}$ be the characteristic function of the Cantor set, $f(x) = 1$ for $x \in \mathcal{C}$ and $f(x) = 0$ for $x \notin \mathcal{C}$. Let $g : [0, 1] \rightarrow \mathbb{R}$ be any bounded function where $\text{Dis}(g) \cap \text{Dis}(\alpha) = \emptyset$.

1. If $\text{Dis}(g)$ is countable, then $g \in \mathcal{R}(\alpha)$.
2. $f \in \mathcal{R}$, but $f \notin \mathcal{R}(\alpha)$.

Is there a general pattern? Define $A \subseteq [a, b]$ *has measure 0* to mean

$$\inf \left\{ \sum_n (b_n - a_n) \mid A \subseteq \bigcup_n (a_n, b_n) \right\} = 0$$

In words: We can cover A by a collection of intervals, the sum of whose lengths can be made arbitrarily small. With this definition, we have:

Theorem 1. If $f : [a, b] \rightarrow \mathbb{R}$ is bounded, then

$$\text{Dis}(f) \text{ has measure } 0 \implies f \in \mathcal{R}$$

This covers our first example as *any countable subset of $[a, b]$ has measure 0*.

We can extend this notion to increasing $\alpha : [a, b] \rightarrow \mathbb{R}$. Say $A \subseteq [a, b]$ *has α -measure 0* if

$$\inf \left\{ \sum_n (\alpha(b_n) - \alpha(a_n)) \mid A \subseteq \bigcup_n (a_n, b_n) \right\} = 0$$

This is the same as measure 0 when α is the identity function. Then we have:

Theorem 2. If $\alpha : [a, b] \rightarrow \mathbb{R}$ is monotonically increasing and $f : [a, b] \rightarrow \mathbb{R}$ is bounded and $\text{Dis}(f) \cap \text{Dis}(\alpha) = \emptyset$, then

$$\text{Dis}(f) \text{ has } \alpha\text{-measure } 0 \implies f \in \mathcal{R}(\alpha)$$

This covers our other two examples. In particular, if α is the Cantor function, then $[0, 1] \setminus \mathcal{C}$ has α -measure 0, or, said differently, $\mathcal{C}$ has α -measure 1.

So any bounded function continuous off the Cantor set is Riemann integrable since the Cantor set has measure 0, but if α is the Cantor function, then the Cantor set has α -measure > 0 and hence f might no longer be in $\mathcal{R}(\alpha)$.

Problem 6.7 (B:6.4:8 – Integrability piece by piece.). Let $s_1 < s_2 < \dots < s_n < \dots$ be elements of an interval $[a, b]$ such that $s_1 = a$ and $\lim_{n \rightarrow \infty} s_n = b$. Let α be an increasing function on $[a, b]$, and let f be any bounded real-valued function on $[a, b]$. For each positive integer i , let f_i denote the restriction of f to $[s_i, s_{i+1}]$, and α_i the restriction of α to that interval.

Prove that $f \in \mathcal{R}(\alpha)$ if and only if for every i , $f_i \in \mathcal{R}(\alpha_i)$, and that when this holds, we have

$$\int_a^b f \, d\alpha = \sum_{i=1}^{\infty} \left(\int_{s_i}^{s_{i+1}} f \, d\alpha \right),$$

with the sum on the right is absolutely convergent. (The i th summand of that sum could, of course, be written more formally as $\int_{s_i}^{s_{i+1}} f_i \, d\alpha_i$.)

Problem 6.8 (B:6.5:2 – Strengthening of 6.16). Show that Theorem 6.16 (p.130) remains valid if the assumption that f is continuous on $[a, b]$ is replaced by the condition that it is bounded, and is continuous at each of the points s_n . Specifically -

We see that no continuity condition is needed until display (24). Say why that display is valid under the condition stated above. Then modify the remainder of the proof (in which Rudin uses integrability of f with respect to α_2 , which under his assumptions follows from Theorem 6.8) to conclude that (23) holds with the upper, respectively the lower integral in place of the left-hand side. Conclude that $f \in \mathcal{R}(\alpha)$ and that (23) holds.

Problem 6.9 (B:6.6:1. – A sort of derivative formula for Riemann-Stieltjes). On $[a, b]$, let α be a strictly increasing function (as defined in **B:5.2:1 (b)**) and f a continuous function, and for $x \in [a, b]$ define $F(x) = \int_a^x f(t) \, d\alpha(t)$.

Show that for all $x \in [a, b]$,

$$\frac{dF(x)}{d\alpha(x)} = f(x),$$

where the left-hand side is defined as $\lim_{t \rightarrow x} (F(x) - F(t)) / (\alpha(x) - \alpha(t))$, and the equality includes the assertion that this limit exists.

Problem 6.10 (B:6.6:5 – Repeated Riemann-Stieltjes integration.). We shall obtain here a generalization of the result of **B:6.6:2** in which ds and dt are replaced by more general expressions $d\alpha(s)$ and $d\beta(t)$.

- (a) Suppose α, β are increasing functions on $[a, b]$ such that $\alpha \in \mathcal{R}(\beta)$, and $f \in \mathcal{R}(\alpha) \cap \mathcal{R}(\beta)$. Show that $\int_a^x f(t) d\alpha(t)$, regarded as a function of x , belongs to $\mathcal{R}(\beta)$, and that

$$\int_{t=a}^b \left(\int_{s=a}^t f(s) d\alpha(s) \right) d\beta(t) = \int_{t=a}^b (\beta(b) - \beta(t)) f(t) d\alpha(t).$$

(Suggestion: First consider the case where f is everywhere > 0 since in this case one can easily describe the least and greatest values of $\int_{s=a}^t f(s) d\alpha(s)$ on any interval as the values at the respective ends of the interval; then get the general case by writing any $f \in \mathcal{R}(\alpha) \cap \mathcal{R}(\beta)$ as a difference of two nonnegative-valued functions in $\mathcal{R}(\alpha) \cap \mathcal{R}(\beta)$. Cf. the method of passing from part **B:6.5:1** (b) to (c).)

- (b) Show that given continuous functions u and v on $[a, b]$, there exists a continuous function w of two variables such that for any $f \in \mathcal{R}$ on $[a, b]$, one has

$$\int_{t=a}^x u(t) \left(\int_{s=a}^t v(s) f(s) ds \right) dt = \int_{t=a}^x w(x, t) f(t) dt.$$

Hint: Prove the case where u and v are nonnegative-valued using (a).

Problem 6.11 (**B:6.8:3** – Integrability of γ' is enough to prove the curve-length formula.). Suppose $\gamma : [a, b] \rightarrow R^k$ is a differentiable curve such that $\gamma' \in \mathcal{R}$. Exercise **B:6.7:1**(a) tells us that given $\epsilon > 0$ we can find a partition $P = \{x_0, \dots, x_n\}$ such that

$$\sum_{i=1}^n \text{diam}(\mathbf{f}([x_{i-1}, x_i])) \Delta x_i < \epsilon.$$

1. Show that if P is such a partition, and we take any point t_i in each interval $[x_{i-1}, x_i]$, then

$$\sum ||\gamma(x_i) - \gamma(x_{i-1})| - |\gamma'(t_i)| \Delta x_i| < \epsilon.$$

2. Deduce that Theorem 6.27 remains true if the assumption " γ' is continuous" is weakened to " $\gamma' \in \mathcal{R}$ ".